

Creating a Complete Solution — Caso # 1

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PARTE 1 — Cargar los datos

Paso 1: Crear archivo Power BI

- Abrir Power BI Desktop
- Crear nuevo archivo
- Guardarlo como: SalesRepAnalysis.pbix

Paso 2: Importar Access

Inicio → Obtener Datos

Seleccionar:

Access Database

Abrir:

SalesRepAnalysis.accdb

Paso 3: Seleccionar tablas

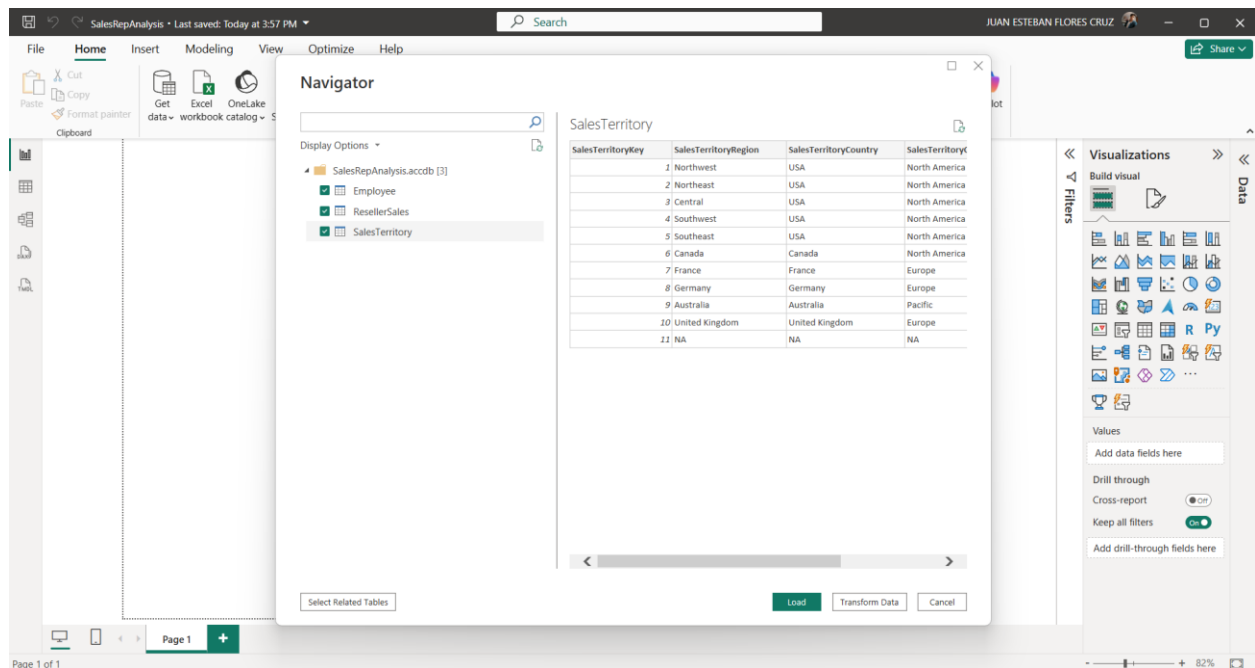
Importa estas tablas:

Tabla	Nombre sugerido
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Employee	Rep
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SalesTerritory	Territory
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ResellerSales	Sales
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PARTE 2 — Transformar datos (Power Query)

Presiona:

Transform Data

Tabla Rep

Deja estas columnas:

- EmployeeKey
- EmployeeNationalID
- FirstName
- LastName
- MiddleName
- Title

Filtrar solo vendedores

En la columna Title:

filtra únicamente los empleados relacionados con ventas

Ejemplo:

Sales Representative

Sales Manager

Crear columna Name

Agregar columna personalizada:

FirstName & " " & LastName

The screenshot shows the Power Query Editor interface. The formula bar at the top displays the custom column formula: `Table.AddColumn(#"Filtered Rows", "Name", each [FirstName] & " " & [LastName])`. Below the formula bar, a table of 14 rows and 7 columns is visible. The columns are: EmployeeNationalID, FirstName, LastName, MiddleName, Title, and the newly added 'Name' column. The 'Name' column contains the concatenated first and last names of the sales representatives. On the right side, the 'Query Settings' pane shows the 'Name' property set to 'Rep' and the 'Applied Steps' list containing 'Source', 'Navigation', 'Removed Columns', 'Filtered Rows', and 'Added Custom'.

	EmployeeNationalID	FirstName	LastName	MiddleName	Title	Name	
1	281	841560125	Michael	Blythe	G	Sales Representative	Michael Blythe
2	282	191644724	Linda	Mitchell	C	Sales Representative	Linda Mitchell
3	283	615389812	Jillian	Carson		Sales Representative	Jillian Carson
4	284	234474252	Garrett	Vargas	R	Sales Representative	Garrett Vargas
5	285	716374314	Tsvi	Reiter	Michael	Sales Representative	Tsvi Reiter
6	286	61161660	Pamela	Anson-Wolfe	O	Sales Representative	Pamela Anson-Wolfe
7	287	139397894	Shu	Ito	K	Sales Representative	Shu Ito
8	288	399771412	José	Saraiva	Edvaldo	Sales Representative	José Saraiva
9	289	987554265	David	Campbell	R	Sales Representative	David Campbell
10	291	668991357	Jae	Pak	B	Sales Representative	Jae Pak
11	292	134219713	Ranjit	Varkey Chudukatil	R	Sales Representative	Ranjit Varkey Chudukatil
12	293	90836195	Tete	Mensa-Annan	A	Sales Representative	Tete Mensa-Annan
13	295	954276278	Rachel	Valdez	B	Sales Representative	Rachel Valdez
14	296	758596752	Lynn	Tsoflas	N	Sales Representative	Lynn Tsoflas

Tabla Territory

Deja: Columnas

- SalesTerritoryKey
- SalesTerritoryRegion
- SalesTerritoryCountry
- SalesTerritoryGroup

Tabla Sales

Deja:

- OrderDateKey
- EmployeeKey
- SalesTerritoryKey
- SalesAmount

Convertir fecha

La columna: OrderDateKey

viene así:

20240115

Cambiar tipo

Selecciona OrderDateKey

Cambiar tipo → Texto

Crear OrderDate

Agregar columna personalizada:

Date.FromText(Text.Start([OrderDateKey],4) & "-" &

Text.Middle([OrderDateKey],4,2) & "-" &

Text.End([OrderDateKey],2))

Nombre: **OrderDate**

Table with 5 columns: OrderDateKey, EmployeeKey, SalesTerritoryKey, SalesAmount, OrderDate. The table contains 28 rows of data. The OrderDate column is highlighted in green. The formula bar shows the DAX formula: = Table.AddColumn(#"Changed Type", "OrderDate", each Date.FromText(Text.Start([OrderDateKey],4) & "-" & Text.Middle([OrderDateKey],4,2) & "-" & Text.End([OrderDateKey],2))). The right sidebar shows the 'APPLIED STEPS' pane with 'Added Custom' selected.

Crear Year y Quarter

Agregar:

- Año

Date.Year([OrderDate])

- Quarter

"Q" & Number.ToText(Date.QuarterOfYear([OrderDate]))

SalesRepAnalysis

Queries [3]

Rep

Sales

Territory

Table.AddColumn(#"Added Custom1", "Quarter", each "Q" & Number.ToText(Date.QuarterOfYear([OrderDate])))

	EmployeeKey	SalesTerritoryKey	SalesAmount	OrderDate	Year	Quarter
1	285	5	2,024.99	01/07/2013	2013	Q3
2	285	5	6,074.98	01/07/2013	2013	Q3
3	285	5	2,024.99	01/07/2013	2013	Q3
4	285	5	2,039.99	01/07/2013	2013	Q3
5	285	5	2,039.99	01/07/2013	2013	Q3
6	285	5	4,079.99	01/07/2013	2013	Q3
7	285	5	2,039.99	01/07/2013	2013	Q3
8	285	5	86.52	01/07/2013	2013	Q3
9	285	5	28.84	01/07/2013	2013	Q3
10	285	5	34.20	01/07/2013	2013	Q3
11	285	5	10.37	01/07/2013	2013	Q3
12	285	5	80.75	01/07/2013	2013	Q3
13	285	5	419.46	01/07/2013	2013	Q3
14	285	5	874.79	01/07/2013	2013	Q3
15	288	6	809.76	01/07/2013	2013	Q3
16	288	6	714.70	01/07/2013	2013	Q3
17	288	6	1,429.41	01/07/2013	2013	Q3
18	288	6	20.75	01/07/2013	2013	Q3
19	288	6	115.36	01/07/2013	2013	Q3
20	288	6	1,445.19	01/07/2013	2013	Q3
21	288	6	6,074.98	01/07/2013	2013	Q3
22	---	---	---	---	---	---

20130701

7 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows

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Agrupar ventas

Debes agrupar por:

- EmployeeKey
- SalesTerritoryKey
- Year
- Quarter

y sumar: SalesAmount

Group By

Specify the columns to group by and one or more outputs.

☐ Basic ☒ Advanced

EmployeeKey

SalesTerritoryKey

Year

Quarter

Add grouping

New column name

Sales Amount

Operation

Sum

Column

SalesAmount

Add aggregation

OK

Cancel

Crear TimePeriod

Nueva columna: Year & Quarter

Ejemplo: 2024Q1

Luego elimina:

- Year
- Quarter

The screenshot shows the Power Query Editor interface. The main area displays a table with the following columns: EmployeeKey, SalesTerritoryKey, Sales Amount, and TimePeriod. The TimePeriod column contains values such as 2013Q3 and 2013Q4. The interface includes a ribbon with various transformation options and a right-hand pane for query settings.

EmployeeKey	SalesTerritoryKey	Sales Amount	TimePeriod
285	5	544530.0264	2013Q3
288	6	428607.3466	2013Q3
282	4	302271.1838	2013Q3
286	1	191328.4939	2013Q3
289	1	203348.2389	2013Q3
283	3	239893.6352	2013Q3
281	3	52539.7917	2013Q3
281	2	217470.9457	2013Q3
281	5	34744.8652	2013Q3
284	6	207335.5039	2013Q3
287	4	278285.4919	2013Q3
283	4	215191.6908	2013Q3
272	1	20544.7015	2013Q3
282	1	113907.9411	2013Q3
287	3	9398.8972	2013Q3
287	1	89317.0089	2013Q3
282	3	42878.212	2013Q3
272	6	2039.994	2013Q3
281	2	351074.5697	2013Q4
282	4	418002.2072	2013Q4
288	6	610342.1933	2013Q4
289	1	303222.1451	2013Q4
286	1	416089.2534	2013Q4
283	3	460772.0749	2013Q4
281	3	64588.6901	2013Q4
287	4	344055.8239	2013Q4
281	5	31322.243	2013Q4
284	6	263094.4185	2013Q4

PARTE 3 — Importar cuotas

Abrir Excel

Importa:

- SalesQuotas.xlsx

Selecciona todas las hojas.

Table: TransformColumnTypes(#"Promoted Headers",{"SalesPerson", type text}, {"Q1", Int64.Type}, {"Q2", Int64.Type})

	SalesPerson	Q1	Q2
1	Amy Alberts	92800	800
2	David Campbell	187200	322400
3	Garrett Vargas	224000	312000
4	Jae Pak	706400	1063200
5	Jillian Carson	571200	757600
6	José Saraiva	455200	664000
7	Linda Mitchell	715200	899200
8	Lynn Tsollias	319200	336800
9	Michael Blythe	679200	695200
10	Pamela Ansman-Wolfe	274400	312000
11	Rachel Valdez	292800	501600
12	Ranjit Varkey Chudukatil	565600	726400
13	Shu-Itto	491200	579200
14	Stephen Jiang	67200	149600
15	Syed Abbas	5600	20800
16	Tete Mensa-Annan	363200	397600
17	Tsvi Reiter	430400	546800

3 COLUMNS, 17 ROWS Column profiling based on top 1000 rows

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Agregar columna Year

Cada hoja representa un año.

Debes agregar una columna:

- **Year**

Unir tablas

Usa: Append Queries

Nombre final: Quotas

Ordenar columnas

Debe quedar:

- SalesPerson
- Year
- Q1
- Q2
- Q3
- Q4

SalesRepAnalysis

Archive Home Transform Add Column View Tools Help

Close & Apply New Source Recent Enter Data Data source settings Manage Parameters Export query results Refresh Preview Advanced Editor Choose Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Combine

Queries [8]

Rep Sales Territory 2013 2014 2015 2016 Quotas

Table.Combine({#"2013", #"2014", #"2015", #"2016"})

	ABC 123 Q1	ABC 123 Q2	ABC 123 Q3	ABC 123 Q4	ABC 123 Year
1 David Campbell	null	null	180800	282400	2013
2 Garrett Vargas	null	null	195200	284800	2013
3 Jillian Carson	null	null	452000	697600	2013
4 José Saraiva	null	null	420000	613600	2013
5 Linda Mitchell	null	null	509600	624800	2013
6 Michael Blythe	null	null	293600	444800	2013
7 Pamela Annsman-Wolfe	null	null	132000	375200	2013
8 Shu Ito	null	null	368000	439200	2013
9 Stephen Jiang	null	null	22400	5600	2013
10 Tsvi Reiter	null	null	535200	733600	2013
11 Amy Alberts	null	null	74400	12000	2014
12 David Campbell	224000	213600	386400	272800	2014
13 Garrett Vargas	212800	272800	433600	263200	2014
14 Jae Pak	null	null	1518400	1009600	2014
15 Jillian Carson	676800	761600	1280000	1081600	2014
16 José Saraiva	466400	466400	396800	361600	2014
17 Linda Mitchell	932000	511200	1084000	807200	2014
18 Michael Blythe	401600	440000	1143200	1059200	2014
19 Pamela Annsman-Wolfe	211200	240800	168000	180800	2014
20 Ranjit Varkey Chudukatil	null	null	474400	327200	2014
21 Shu Ito	378400	394400	827200	650400	2014
22 Stephen Jiang	72800	112000	56000	123200	2014
23 Tete Mensa-Annan	null	null	null	256800	2014
24 Tsvi Reiter	624000	641600	587200	517600	2014
25 Amy Alberts	26400	88000	147200	259200	2015
26 David Campbell	120800	227200	504800	297600	2015
27 Garrett Vargas	191200	334400	405600	362400	2015
28 Jae Pak	750400	1023200	1204800	1135200	2015

6 COLUMNS, 58 ROWS Column profiling based on top 1000 rows

Query Settings

PROPERTIES Name Quotas All Properties

APPLIED STEPS Source

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Hacer Unpivot

Selecciona:

- SalesPerson
- Year

Luego:

Transform → Unpivot Other Columns

Renombrar

Antes: Attribute

Después: Quarter

Value: Quota

SalesRepAnalysis

Queries [8]

Rep
Sales
Territory
2013
2014
2015
2016
Quotas

Table

fx = Table.RenameColumns(#Unpivoted Other Columns,{"Attribute", "Quarter"}, {"Value", "Quota"})

Rep	Year	Quarter	Quota
1 David Campbell	2013	Q3	180800
2 David Campbell	2013	Q4	282400
3 Garrett Vargas	2013	Q3	195200
4 Garrett Vargas	2013	Q4	284800
5 Jillian Carson	2013	Q3	452000
6 Jillian Carson	2013	Q4	697600
7 José Saraiva	2013	Q3	420000
8 José Saraiva	2013	Q4	613600
9 Linda Mitchell	2013	Q3	509600
10 Linda Mitchell	2013	Q4	624800
11 Michael Blythe	2013	Q3	293600
12 Michael Blythe	2013	Q4	444800
13 Pamela Ansman-Wolfe	2013	Q3	132000
14 Pamela Ansman-Wolfe	2013	Q4	375200
15 Shu Ito	2013	Q3	368000
16 Shu Ito	2013	Q4	439200
17 Stephen Jiang	2013	Q3	22400
18 Stephen Jiang	2013	Q4	5600
19 Tsvi Reiter	2013	Q3	535200
20 Tsvi Reiter	2013	Q4	733600
21 Amy Alberts	2014	Q3	74400
22 Amy Alberts	2014	Q4	12000
23 David Campbell	2014	Q1	224000
24 David Campbell	2014	Q2	213600
25 David Campbell	2014	Q3	386400
26 David Campbell	2014	Q4	272800
27 Garrett Vargas	2014	Q1	212800
28 Garrett Vargas	2014	Q2	272800

4 COLUMNS, 163 ROWS Column profiling based on top 1000 rows

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Crear TimePeriod

Year & Quarter

Ejemplo: 2024Q2

Luego elimina:

- Year
- Quarter

SalesRepAnalysis

Queries [8]

Rep
Sales
Territory
2013
2014
2015
2016
Quotas

Table

fx = Table.RemoveColumns(#Added Custom,{"Year", "Quarter"})

Rep	Quota	TimePeriod
1 David Campbell	180800	2013Q3
2 David Campbell	282400	2013Q4
3 Garrett Vargas	195200	2013Q3
4 Garrett Vargas	284800	2013Q4
5 Jillian Carson	452000	2013Q3
6 Jillian Carson	697600	2013Q4
7 José Saraiva	420000	2013Q3
8 José Saraiva	613600	2013Q4
9 Linda Mitchell	509600	2013Q3
10 Linda Mitchell	624800	2013Q4
11 Michael Blythe	293600	2013Q3
12 Michael Blythe	444800	2013Q4
13 Pamela Ansman-Wolfe	132000	2013Q3
14 Pamela Ansman-Wolfe	375200	2013Q4
15 Shu Ito	368000	2013Q3
16 Shu Ito	439200	2013Q4
17 Stephen Jiang	22400	2013Q3
18 Stephen Jiang	5600	2013Q4
19 Tsvi Reiter	535200	2013Q3
20 Tsvi Reiter	733600	2013Q4
21 Amy Alberts	74400	2014Q3
22 Amy Alberts	12000	2014Q4
23 David Campbell	224000	2014Q1
24 David Campbell	213600	2014Q2
25 David Campbell	386400	2014Q3
26 David Campbell	272800	2014Q4
27 Garrett Vargas	212800	2014Q1
28 Garrett Vargas	272800	2014Q2

3 COLUMNS, 163 ROWS Column profiling based on top 1000 rows

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PARTE 4 — Crear tabla Period

Duplicar Quotas

- Click derecho en Quotas
- Duplicate

Nombre: Period

Quita:

- SalesPerson
- Quota
- Quitar duplicados

Selecciona:

- TimePeriod

Luego:

- Remove Duplicates

The screenshot shows the Power BI Desktop interface with the 'SalesRepAnalysis' query loaded. The 'Period' column is selected in the 'TimePeriod' table. The 'Remove Duplicates' step is highlighted in the 'APPLIED STEPS' pane. The formula bar shows '= Table.Distinct("#Removed Columns")'.

Rep	Sales	Territory	2013	2014	2015	2016	Quotas	Period
1	2013Q3							
2	2013Q4							
3	2014Q3							
4	2014Q4							
5	2014Q1							
6	2014Q2							
7	2015Q1							
8	2015Q2							
9	2015Q3							
10	2015Q4							
11	2016Q1							
12	2016Q2							

PARTE 5 — Crear relaciones

Ir a:

Model View

Relaciones

Rep ↔ Sales

EmployeeKey

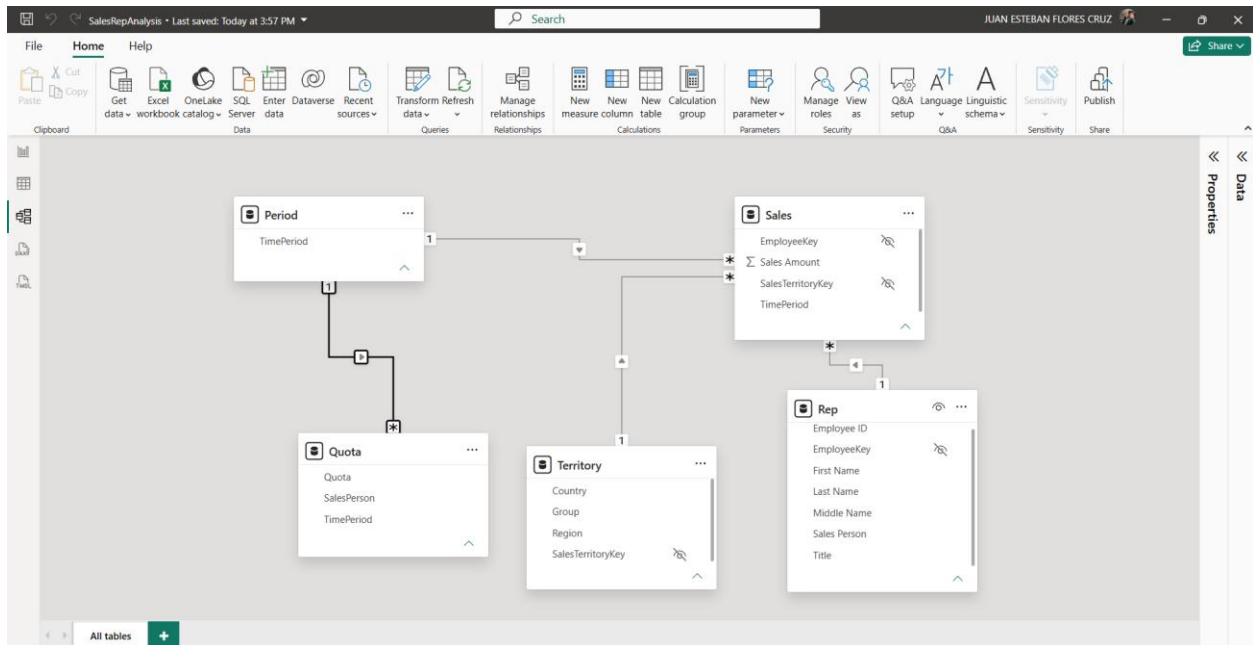
Territory ↔ Sales

SalesTerritoryKey

Period ↔ Sales

TimePeriod

Period ↔ Quotas



◆ PARTE 6 — Crear medidas DAX

En tabla Sales:

Total Sales = SUM(Sales[Sales Amount])

En Quotas:

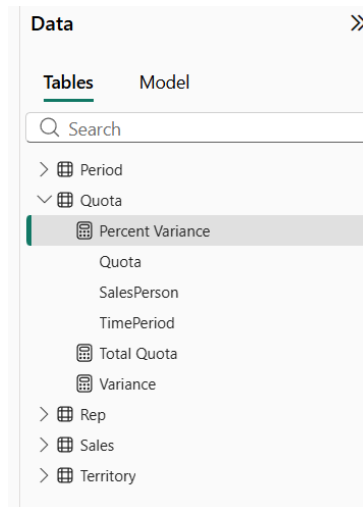
Total Quota = SUM(Quota[Quota])

Variance:

Variance = [Total Sales] - [Total Quota]

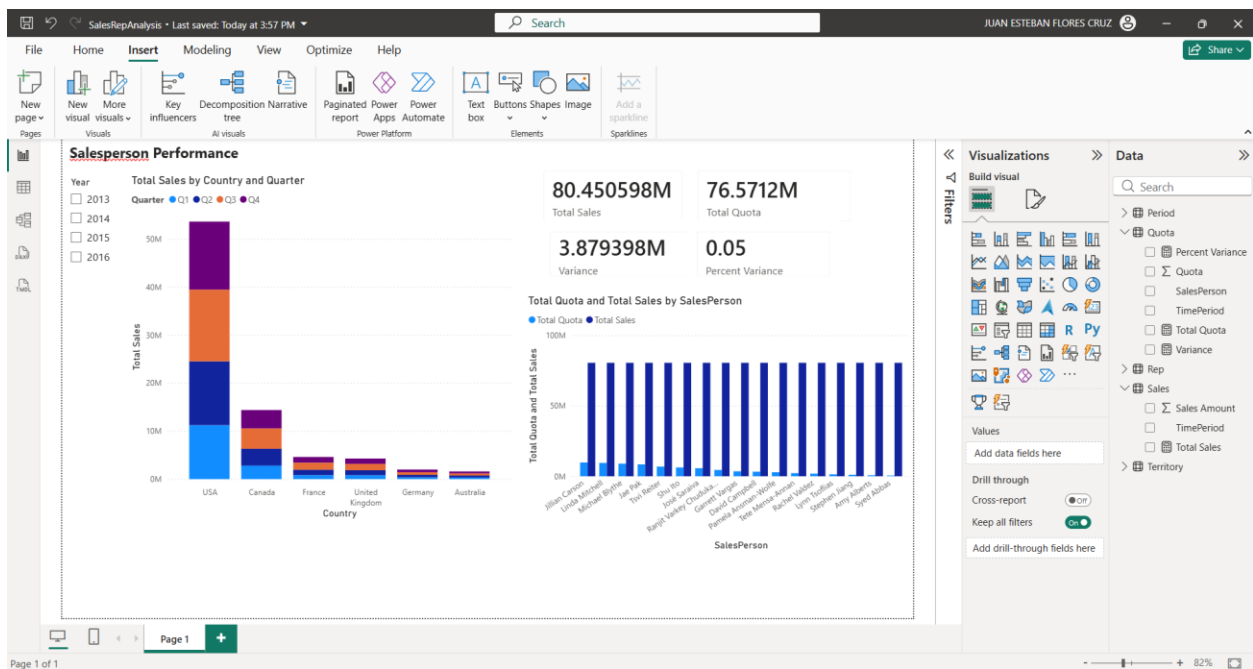
Percent Variance

Percent Variance = DIVIDE([Variance],[Total Quota],BLANK())



PARTE 7 — Crear Dashboard

Página 1



Página 2

